

Taylor-Informed Indirect Adaptive Predictive Control Using Jacobian-Frozen Affine Predictors

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Abstract—This paper develops a Taylor-informed indirect adaptive predictive control framework for nonlinear sampled-data systems using Jacobian-frozen affine predictors. A finite Taylor expansion approximates the sampled nonlinear dynamics, and recursive least squares (RLS) identifies its polynomial coefficients online. At each sampling instant, the Jacobian of the identified map is evaluated at the current operating point and frozen over the prediction horizon, yielding an affine predictor for model predictive control. In contrast to generic nonlinear feature dictionaries, the implemented polynomial dictionary is a forward-Euler/Taylor-structure-informed reduced dictionary. Exact joint-odd symmetry eliminates even-total-degree monomials, whereas additional forward-Euler-informed pruning constitutes a deliberate model reduction. Numerical simulations on an unstable nonlinear benchmark compare different Taylor degrees. The results show that higher-order models improve tracking accuracy as the operating point moves farther from the expansion point while maintaining comparable control effort. The complete MATLAB implementation is publicly available to facilitate reproducibility.

I. INTRODUCTION

Indirect adaptive predictive control combines online identification with model predictive control (MPC), adapting to streaming input–output data while retaining the structure of linear predictive control [1]–[3]. At each sampling instant, the predictive model is updated recursively to compute a finite-horizon control sequence. Recent developments have extended this framework to aerospace and robotic applications [4]–[6].

Adaptive predictive control for nonlinear systems remains an active research topic. Existing approaches include nonlinear model predictive control, lifting based on observable dictionaries, and adaptive predictive methods with nonlinear regressors [7]–[11]. More recently, nonlinear extensions of the predictive cost adaptive control (PCAC) framework [12] have used kernel, polynomial, Fourier, and spline feature dictionaries [13], [14]. These methods replace the linear regressor with a richer feature dictionary whose coefficients are estimated through recursive least squares (RLS). Polynomial, kernel, and other nonlinear observable representations can thus increase expressiveness while preserving parameter linearity [15]–[17]. Although these dictionaries often improve approximation capability, they are often drawn from general basis families whose connection to the underlying nonlinear plant is indirect.

This work takes a different approach. Instead of introducing a generic nonlinear dictionary, we construct a reduced

feature set from the Taylor structure of the sampled-data dynamics and structural information from a forward-Euler expansion. The resulting monomials have a clear analytical interpretation. Exact symmetries of the sampled flow justify some eliminations, while further omissions are imposed deliberately as forward-Euler-informed model reduction. This additional reduction does not imply that every omitted monomial vanishes in the sampled flow.

An exact nonlinear basis can be used directly when it is known and trusted. In many applications, however, only partial structural knowledge is available. A Taylor dictionary then provides a common local representation without committing to a particular functional basis. Increasing its degree accommodates richer local nonlinear behavior through coefficient adaptation rather than regressor redesign.

After RLS identifies the polynomial coefficients online, the sampled nonlinear map is converted into a locally affine predictive model. Its Jacobian is evaluated at the current operating point, and the resulting affine coefficients are frozen over the prediction horizon. This construction avoids repeated nonlinear propagation while retaining online adaptation through the identified Taylor coefficients. The predictor is then used in a conventional MPC formulation.

Through numerical simulation, this paper investigates how the Taylor degree affects closed-loop tracking. Structured Taylor dictionaries of different orders are compared on a nonlinear benchmark to evaluate the trade-off between model complexity and control performance. To facilitate reproducibility, the complete MATLAB implementation used throughout this study is publicly available at <https://github.com/tamwng/taylor-informed-iapc>.

II. SYSTEM DESCRIPTION

Consider the continuous-time nonlinear system

$$\dot{x} = f(x, u), \quad (1)$$

where $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$, $x \in \mathbb{R}^n$ is the state, and $u \in \mathbb{R}^m$ is the control input.

Under zero-order hold with sampling period $T_s > 0$,

$$u(t) = u_k, \quad t \in [kT_s, (k+1)T_s),$$

where $t \in \mathbb{R}_{\geq 0}$ denotes continuous time and $k \in \mathbb{N}_0$ is the sampling index. The exact sampled-data dynamics are

$$x_{k+1} = F(x_k, u_k), \quad (2)$$

where $F : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ denotes the sampled-data flow map associated with (1).

Select an equilibrium pair (x_e, u_e) satisfying

$$f(x_e, u_e) = 0. \quad (3)$$

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Then,

$$F(x_e, u_e) = x_e.$$

Define the shifted variables

$$\tilde{x} := x - x_e, \quad \tilde{u} := u - u_e.$$

The shifted sampled-data dynamics are

$$\tilde{x}_{k+1} = \bar{F}(\tilde{x}_k, \tilde{u}_k), \quad (4)$$

where

$$\bar{F}(\tilde{x}, \tilde{u}) := F(x_e + \tilde{x}, u_e + \tilde{u}) - x_e. \quad (5)$$

By construction,

$$\bar{F}(0, 0) = 0. \quad (6)$$

The following assumptions are used throughout this paper.

Assumption 1. The state x_k is available for feedback at each time k .

Assumption 2. $\bar{F}$ is well-defined on a neighborhood $\mathcal{N} \subset \mathbb{R}^{n+m}$ of the origin.

Assumption 3. $\bar{F} : \mathcal{N} \rightarrow \mathbb{R}^n$ is of class $\mathcal{C}^{D+1}$ on $\mathcal{N}$.

III. TAYLOR APPROXIMATION OF THE SAMPLED DYNAMICS

Under Assumption 3, the map $\bar{F} : \mathcal{N} \rightarrow \mathbb{R}^n$ admits a finite Taylor approximation about the origin.

Let

$$p := n + m, \quad z := \begin{bmatrix} \tilde{x} \\ \tilde{u} \end{bmatrix} \in \mathbb{R}^p. \quad (7)$$

Since $\bar{F}(0) = 0$, the order- D Taylor approximation is

$$\bar{F}(z) = Mz + \sum_{d=2}^D F_d(z) + r_D(z), \quad (8)$$

where

$$M := \left. \frac{\partial \bar{F}}{\partial z} \right|_{z=0} \in \mathbb{R}^{n \times p}, \quad (9)$$

and $r_D : \mathcal{N} \rightarrow \mathbb{R}^n$ is the Taylor remainder. For each $d \in \{2, \dots, D\}$, $F_d : \mathbb{R}^p \rightarrow \mathbb{R}^n$ collects the homogeneous polynomial terms of total degree d . Thus, F_2 contains all quadratic terms, F_3 contains all cubic terms, and so forth. For $D = 1$, the sum in (8) is empty.

The local Taylor approximation in (8) is a polynomial in the components of z . Define the full Taylor index set

$$\mathcal{A}_D := \{\alpha \in \mathbb{N}_0^p : 1 \leq |\alpha| \leq D\}, \quad |\alpha| := \sum_{j=1}^p \alpha_j. \quad (10)$$

For each $\alpha \in \mathcal{A}_D$, define the monomial

$$z^\alpha := \prod_{j=1}^p z_j^{\alpha_j}, \quad (11)$$

where z_j is the j -th component of z .

Let $\mathcal{A}_D = \{\alpha_1, \dots, \alpha_q\}$. The polynomial regressor is

$$\phi_D(z) := [z^{\alpha_1} \quad \dots \quad z^{\alpha_q}]^\top \in \mathbb{R}^q. \quad (12)$$

The regressor ϕ_D contains every pure and crossed monomial of total degree from 1 through D . Therefore,

$$q = \binom{p+D}{D} - 1. \quad (13)$$

The order- D Taylor model can therefore be written compactly as

$$\bar{F}(z) = \Theta \phi_D(z) + r_D(z), \quad (14)$$

where $\Theta \in \mathbb{R}^{n \times q}$ contains the Taylor coefficients.

IV. RECURSIVE TAYLOR COEFFICIENT IDENTIFICATION

The Taylor model (14) is linear in the coefficient matrix Θ . After neglecting the Taylor remainder, identification of the sampled dynamics reduces to estimation of Θ .

For each $k \geq 0$, define

$$z_k := \begin{bmatrix} \tilde{x}_k \\ \tilde{u}_k \end{bmatrix}, \quad \phi_k := \phi_D(z_k). \quad (15)$$

The a priori one-step prediction of $\tilde{x}_{k+1}$ is

$$\hat{\tilde{x}}_{k+1} = \hat{\Theta}_k \phi_k. \quad (16)$$

Vectorize the coefficient matrix as

$$\theta_k := \text{vec}(\hat{\Theta}_k) \in \mathbb{R}^{nq}, \quad (17)$$

and define the regression matrix

$$\varphi_k := \phi_k^\top \otimes I_n \in \mathbb{R}^{n \times nq}. \quad (18)$$

Then (16) becomes

$$\hat{\tilde{x}}_{k+1} = \varphi_k \theta_k. \quad (19)$$

The prediction error is

$$e_k := \tilde{x}_{k+1} - \varphi_k \theta_k. \quad (20)$$

Using standard RLS with forgetting factor $\lambda \in (0, 1]$ and $P_0 \succ 0$, define

$$L_k = \lambda^{-1} P_k, \quad (21)$$

$$P_{k+1} = L_k - L_k \varphi_k^\top (I_n + \varphi_k L_k \varphi_k^\top)^{-1} \varphi_k L_k, \quad (22)$$

$$\theta_{k+1} = \theta_k + P_{k+1} \varphi_k^\top e_k. \quad (23)$$

Processing the measured transition $(z_k, \tilde{x}_{k+1})$ yields the posterior RLS estimate (P_{k+1}, θ_{k+1}) . In the numerical study, these one-step transitions come from the zero-order-hold RK4 simulation of the continuous-time plant; forward Euler does not propagate the simulated plant. Once $\tilde{x}_{k+1}$ is measured, θ_{k+1} is available for the subsequent control synthesis.

Using this posterior estimate, reshape θ_{k+1} into $\hat{\Theta}_{k+1} \in \mathbb{R}^{n \times q}$ and define the identified sampled Taylor map

$$\hat{F}_{k+1}(z) := \hat{\Theta}_{k+1} \phi_D(z). \quad (24)$$

Thus, $\hat{F}_{k+1}$ follows from (14) by neglecting the Taylor remainder and replacing the unknown coefficient matrix Θ with its RLS estimate $\hat{\Theta}_{k+1}$.

The map $\hat{F}_{k+1}$ is nonlinear in z but linear in the identified coefficients, which enables direct RLS identification. Other RLS variants, including variable-rate forgetting or covariance reset, can be used without changing the Jacobian-frozen construction below.

In the sequel, $\hat{F}_k$ denotes the most recent identified map available at the current control instant.

V. JACOBIAN-FROZEN AFFINE PREDICTOR

Because the identified Taylor map $\hat{F}_k$ is nonlinear in the shifted state and input, its exact finite-horizon propagation would lead to nonlinear predictive control. We instead construct a local affine predictor by freezing the Jacobian of $\hat{F}_k$ at the current operating point.

A. Operating Point

At step k , the shifted state $\tilde{x}_k$ is measured, $\tilde{u}_k$ is currently applied, and the control computation produces $\tilde{u}_{k+1}$. Accordingly, the operating point is

$$\bar{z}_k := z_k = \begin{bmatrix} \tilde{x}_k \\ \tilde{u}_k \end{bmatrix} \in \mathbb{R}^{n+m}. \quad (25)$$

The initial input u_0 is prescribed.

The Taylor approximation remains centered at the equilibrium $z = 0$; the operating point $\bar{z}_k$ is used only to evaluate the Jacobian.

B. Jacobian Evaluation

Compute the Jacobian of the identified Taylor map at $\bar{z}_k$:

$$J_k := \left. \frac{\partial \hat{F}_k}{\partial z} \right|_{z=\bar{z}_k} \in \mathbb{R}^{n \times (n+m)}. \quad (26)$$

Partition

$$J_k = [A_k \ B_k], \quad (27)$$

where $A_k \in \mathbb{R}^{n \times n}$ and $B_k \in \mathbb{R}^{n \times m}$ are the Jacobians of $\hat{F}_k$ with respect to the shifted state and input, respectively.

The first-order local approximation of $\hat{F}_k$ at $\bar{z}_k$ is

$$\hat{F}_k(z) \approx \hat{F}_k(\bar{z}_k) + J_k(z - \bar{z}_k). \quad (28)$$

Substituting $z = [\tilde{x}^\top \ \tilde{u}^\top]^\top$ and partitioning J_k according to (27) yields

$$\hat{F}_k(z) \approx c_k + A_k \tilde{x} + B_k \tilde{u}, \quad (29)$$

where

$$c_k := \hat{F}_k(\bar{z}_k) - A_k \tilde{x}_k - B_k \tilde{u}_k. \quad (30)$$

The resulting Jacobian-frozen predictor is

$$\tilde{x}_{i+1|k} = c_k + A_k \tilde{x}_{i|k} + B_k \tilde{u}_{i|k}, \quad i = 0, \dots, N-1. \quad (31)$$

The matrices A_k , B_k , and c_k remain fixed over the prediction horizon.

C. Reduction to Linear PCAC

Linear PCAC is recovered as the first-order special case of the proposed scheme.

Corollary 1: If $D = 1$, then the Jacobian-frozen predictor (31) reduces to the linear PCAC propagation model.

Proof: For $D = 1$, the Taylor regressor satisfies $\phi_1(z) = z$. Hence,

$$\hat{F}_k(z) = \hat{\Theta}_k z, \quad (32)$$

where $\hat{\Theta}_k \in \mathbb{R}^{n \times (n+m)}$. Therefore,

$$J_k = \hat{\Theta}_k = [A_k \ B_k] \quad (33)$$

is independent of the operating point. Moreover,

$$c_k = \hat{F}_k(\bar{z}_k) - J_k \bar{z}_k \quad (34)$$

$$= \hat{\Theta}_k \bar{z}_k - \hat{\Theta}_k \bar{z}_k = 0. \quad (35)$$

Substituting $c_k = 0$ into (31) yields

$$\tilde{x}_{i+1|k} = A_k \tilde{x}_{i|k} + B_k \tilde{u}_{i|k}, \quad (36)$$

which is the linear PCAC propagation model. ■

D. Elimination of Factor-Freezing Ambiguity

Jacobian freezing avoids arbitrary choices when approximating mixed nonlinear terms. For example, consider

$$F(y, u) = \alpha y + \beta u + \alpha u^2 + b u y + c y^2. \quad (37)$$

At $(\bar{y}, \bar{u})$, the Jacobian-frozen predictor is

$$F(y, u) \approx C_{\text{loc}} + A_{\text{loc}} y + B_{\text{loc}} u, \quad (38)$$

where

$$A_{\text{loc}} = \alpha + b\bar{u} + 2c\bar{y}, \quad (39)$$

$$B_{\text{loc}} = \beta + 2a\bar{u} + b\bar{y}, \quad (40)$$

$$C_{\text{loc}} = F(\bar{y}, \bar{u}) - A_{\text{loc}} \bar{y} - B_{\text{loc}} \bar{u}. \quad (41)$$

Thus, the cross term satisfies

$$u y \approx \bar{u} y + \bar{y} u - \bar{u} \bar{y}. \quad (42)$$

This approximation matches both the value and the Jacobian at the operating point without requiring a choice of whether to freeze u or y .

VI. PREDICTIVE CONTROL SYNTHESIS

Following the PCAC formulation [12], the Jacobian-frozen predictor (31) is applied over a prediction horizon N .

At step k , initialize

$$\tilde{x}_{0|k} = \tilde{x}_k, \quad \tilde{u}_{0|k} = \tilde{u}_k. \quad (43)$$

The predicted states satisfy

$$\tilde{x}_{i+1|k} = c_k + A_k \tilde{x}_{i|k} + B_k \tilde{u}_{i|k}, \quad i = 0, \dots, N-1. \quad (44)$$

Here, $\tilde{u}_{0|k}$ is known, whereas $\tilde{u}_{1|k}, \dots, \tilde{u}_{N-1|k}$ are decision variables. This indexing follows the PCAC convention: $\tilde{u}_{1|k}$ is applied at step $k+1$.

Let

$$\Delta \tilde{u}_{i|k} := \tilde{u}_{i|k} - \tilde{u}_{i-1|k}, \quad i = 1, \dots, N-1, \quad (45)$$

and let $\tilde{r}_{i|k}$ denote the reference in shifted coordinates.

Let the state and input constraint sets be the convex polyhedra

$$\mathcal{X} = \{\tilde{x} \in \mathbb{R}^n : H_x \tilde{x} \leq h_x\}, \quad (46)$$

$$\mathcal{U} = \{\tilde{u} \in \mathbb{R}^m : H_u \tilde{u} \leq h_u\}. \quad (47)$$

To soften the state constraints, introduce $\varepsilon_{i|k} \geq 0$ and solve

$$\begin{aligned} & \underset{\substack{\tilde{x}_{1|k}, \dots, \tilde{x}_{N|k}, \\ \tilde{u}_{1|k}, \dots, \tilde{u}_{N-1|k}, \\ \varepsilon_{1|k}, \dots, \varepsilon_{N|k}}}{\text{minimize}} & \frac{1}{2} \sum_{i=1}^N \|\tilde{x}_{i|k} - \tilde{r}_{i|k}\|_{Q_i}^2 \\ & + \frac{1}{2} \sum_{i=1}^{N-1} \|\Delta \tilde{u}_{i|k}\|_R^2 + \frac{1}{2} \sum_{i=1}^N \|\varepsilon_{i|k}\|_S^2 \end{aligned} \quad (48)$$

subject to (44),

$$H_x \tilde{x}_{i|k} \leq h_x + \varepsilon_{i|k}, \quad i = 1, \dots, N,$$

$$H_u \tilde{u}_{i|k} \leq h_u, \quad i = 1, \dots, N-1,$$

$$\varepsilon_{i|k} \geq 0, \quad i = 1, \dots, N,$$

where $Q_i \succeq 0$, $R \succ 0$, $S \succ 0$, and $\|v\|_Q^2 := v^\top Q v$.

Because A_k , B_k , and c_k are fixed over the horizon, (48) is a convex quadratic program. The slack variables prevent infeasibility caused solely by the state constraints but permit violation of the nominal state constraint. No recursive-feasibility or closed-loop stability guarantee is asserted because the identified model and its Jacobian-frozen predictor vary online.

Let $\tilde{u}_{1|k}^*$ denote the first optimized input. The computed control is

$$u_{k+1} = u_e + \tilde{u}_{1|k}^*. \quad (49)$$

Thus, u_k generates the measured transition from k to $k+1$, while the newly computed u_{k+1} is held over the subsequent sampling interval.

VII. CASE STUDY: UNSTABLE TRIGONOMETRIC PLANT

A. Plant and Sampled Dynamics

Consider the continuous-time scalar system

$$\dot{x} = f(x, u) := a \sin x \cos x + b \cos^2 x \sin u, \quad (50)$$

where $a > 0$ and $b > 0$. We select $(x_e, u_e) = (0, 0)$ as the equilibrium. Hence, $\tilde{x} = x$ and $\tilde{u} = u$, and tildes are omitted throughout this section. Near the origin,

$$\dot{x} = ax + bu + \mathcal{O}(\|(x, u)\|^3), \quad (51)$$

so the equilibrium is open-loop unstable and its linearization is controllable.

Under zero-order hold with sampling period $T_s = 0.05$ s, let F denote the exact sampled-data flow map. In the numerical study, an RK4 step propagates the continuous-time plant over each sampling interval under a constant input. The RLS estimator identifies the resulting one-step transitions; forward Euler does not propagate the simulated plant.

For analytical reference only, applying a forward-Euler step to (50) gives

$$\begin{aligned} x_{k+1} &= F_{\text{FE}}(x_k, u_k) \\ &:= x_k + T_s (a \sin x_k \cos x_k + b \cos^2 x_k \sin u_k). \end{aligned} \quad (52)$$

Its linear expansion near the origin is

$$x_{k+1} = (1 + aT_s)x_k + bT_s u_k + \mathcal{O}(\|(x_k, u_k)\|^3), \quad (53)$$

whose open-loop eigenvalue is $1 + aT_s > 1$.

Since

$$f(-x, -u) = -f(x, u), \quad (54)$$

uniqueness of solutions implies the exact joint-odd symmetry of the sampled flow,

$$F(-x, -u) = -F(x, u).$$

The identity also holds exactly for the implemented RK4 map because every stage preserves the joint sign reversal. Consequently, neither the sampled flow nor the RK4 map has even-total-degree terms in its Taylor expansion about the origin.

B. Taylor Structure and Coefficient Pruning

Let

$$z = [x \ u]^\top. \quad (55)$$

As an analytical reference, expanding the forward-Euler expression (52) about the origin through degree five yields

$$\begin{aligned} F_{\text{FE}}(x, u) &= (1 + aT_s)x + bT_s u \\ &\quad - \frac{2aT_s}{3}x^3 - bT_s x^2 u - \frac{bT_s}{6}u^3 \\ &\quad + \frac{2aT_s}{15}x^5 + \frac{bT_s}{3}x^4 u + \frac{bT_s}{6}x^2 u^3 + \frac{bT_s}{120}u^5 \\ &\quad + \mathcal{O}(\|z\|^7). \end{aligned} \quad (56)$$

Joint-odd symmetry removes every even-total-degree term exactly. The implemented dictionary then applies a deliberate additional model reduction informed by the forward-Euler structure. For a degree- D truncation, it retains only monomials of the form

$$x^{2i+1}, \quad x^{2i}u^{2j+1}, \quad i, j \in \mathbb{N}_0, \quad (57)$$

with total degree at most D . This additional pruning does not imply that every omitted odd-total-degree monomial vanishes in the RK4 sampled flow. The resulting feature set is therefore called a forward-Euler/Taylor-structure-informed reduced dictionary.

TABLE I
NUMBER OF ESTIMATED COEFFICIENTS IN THE FULL AND IMPLEMENTED REDUCED DICTIONARIES.

Degree D	Full dictionary	Reduced dictionary
1	2	2
3	9	5
5	20	9
7	35	14

The resulting regressors are

$$\begin{aligned} \varphi_1^{\text{FE}} &= [x \ u]^\top, \\ \varphi_3^{\text{FE}} &= [x \ u \ x^3 \ x^2 u \ u^3]^\top, \\ \varphi_5^{\text{FE}} &= [x \ u \ x^3 \ x^2 u \ u^3 \ x^5 \ x^4 u \ x^2 u^3 \ u^5]^\top, \\ \varphi_7^{\text{FE}} &= [(\varphi_5^{\text{FE}})^\top \ x^7 \ x^6 u \ x^4 u^3 \ x^2 u^5 \ u^7]^\top. \end{aligned}$$

Table I compares the coefficient counts of the full Taylor dictionary and its implemented reduction.

For analytical reference, (56) gives the coefficient vector associated with φ_5^{FE} as

$$\theta_5^* = \begin{bmatrix} 1+aT_s \\ bT_s \\ -2aT_s/3 \\ -bT_s \\ -bT_s/6 \\ 2aT_s/15 \\ bT_s/3 \\ bT_s/6 \\ bT_s/120 \end{bmatrix}. \quad (58)$$

The forward-Euler-informed model reduction therefore decreases the number of estimated coefficients from 20 to 9 for $D = 5$ and from 35 to 14 for $D = 7$. The vector θ_5^* is only an analytical forward-Euler reference: the controller neither uses it nor compares it numerically with the identified RK4-transition coefficients. Instead, RLS estimates the retained coefficients online.

No constant monomial is estimated. The affine term arises subsequently from Jacobian freezing.

C. Identified Map and Jacobian-Frozen Predictor

For each Taylor degree $D \in \{1, 3, 5, 7\}$, let φ_D^{FE} denote the implemented reduced regressor defined by the preceding monomial pattern. The identified sampled map is

$$\hat{F}_{D,k}(x, u) = \hat{\theta}_{D,k}^\top \varphi_D^{\text{FE}}(x, u), \quad (59)$$

where $\hat{\theta}_{D,k}$ is updated online using RLS.

At each step k , the Jacobian-frozen coefficients are evaluated at the current operating point (x_k, u_k) :

$$A_k = \left. \frac{\partial \hat{F}_{D,k}}{\partial x} \right|_{(x_k, u_k)}, \quad (60)$$

$$B_k = \left. \frac{\partial \hat{F}_{D,k}}{\partial u} \right|_{(x_k, u_k)}, \quad (61)$$

$$c_k = \hat{F}_{D,k}(x_k, u_k) - A_k x_k - B_k u_k. \quad (62)$$

For the fifth-order reduced regressor φ_5^{FE} , let $\hat{\theta}_{ij,k}$ denote the estimated coefficient multiplying $x^i u^j$. The resulting Jacobian coefficients are

$$\begin{aligned} A_k &= \hat{\theta}_{10,k} + 3\hat{\theta}_{30,k}x_k^2 + 2\hat{\theta}_{21,k}x_k u_k \\ &\quad + 5\hat{\theta}_{50,k}x_k^4 + 4\hat{\theta}_{41,k}x_k^3 u_k + 2\hat{\theta}_{23,k}x_k u_k^3, \end{aligned} \quad (63)$$

$$\begin{aligned} B_k &= \hat{\theta}_{01,k} + \hat{\theta}_{21,k}x_k^2 + 3\hat{\theta}_{03,k}u_k^2 \\ &\quad + \hat{\theta}_{41,k}x_k^4 + 3\hat{\theta}_{23,k}x_k^2 u_k^2 + 5\hat{\theta}_{05,k}u_k^4. \end{aligned} \quad (64)$$

TABLE II
NUMERICAL CONFIGURATION.

Quantity	Value
Plant parameters	$a = 0.5, b = 1.5$
Sampling period	$T_s = 0.05$ s
Plant integration	RK4 with ZOH input
Initial state	$x_0 = 0$
Constraints	soft $ x_k \leq 0.75$, hard $ u_k \leq 1$
Taylor degrees	$D \in \{1, 3, 5, 7\}$
Prediction horizon	$N = 8$
Forgetting factor	$\lambda = 1$
Initial RLS estimate	$\hat{\theta}_{D,0} = 0_{q_D^{\text{FE}}}$
Initial covariance	$P_0 = 10^5 I_{q_D^{\text{FE}}}$
State weights	$Q_i = 1, i < N, Q_N = 10$
Input-increment weight	$R = 5 \times 10^{-2}$
State-slack weight	$S = 10^5$
Initialization length	$N_{\text{id}} = 60$ samples
Initialization input	$u_k \in \{0, \pm 0.15, \pm 0.30, \pm 0.45, \pm 0.60\}$
Reference type	Amplitude-swept sinusoid
Segment length	$K_r = 300$ samples
Sinusoidal frequency	0.10 Hz
Amplitudes	$\{0.25, 0.55, 0.80, 0.90, 0\}$

The resulting Jacobian-frozen predictor is

$$x_{i+1|k} = c_k + A_k x_{i|k} + B_k u_{i|k}, \quad i = 0, \dots, N-1. \quad (65)$$

For analytical reference only, the Jacobian of the forward-Euler expression (52) is

$$A_{\text{FE}}(x, u) = 1 + T_s [a \cos(2x) - b \sin(2x) \sin u], \quad (66)$$

$$B_{\text{FE}}(x, u) = T_s b \cos^2 x \cos u. \quad (67)$$

VIII. NUMERICAL EVALUATION

This section evaluates how the Taylor degree affects closed-loop tracking. The continuous-time plant (50) is simulated under zero-order hold using fourth-order Runge-Kutta (RK4) with $T_s = 0.05$ s over each sampling interval. RLS identifies the one-step transitions generated by this RK4 simulation. The controller combines the identified forward-Euler/Taylor-structure-informed reduced map with the Jacobian-frozen predictor developed above. All cases $D \in \{1, 3, 5, 7\}$ use the same initialization data, constraints, prediction horizon, and control weights.

The complete MATLAB implementation used throughout this study is publicly available at <https://github.com/tamwng/taylor-informed-iapc>.

A. Simulation Setup

The numerical configuration is summarized in Table II. During the first N_{id} samples, a fixed multilevel input sequence is applied while the RLS estimator is updated. These initialization levels excite the odd monomials retained in the reduced dictionary. The predictive controller is activated at $k = N_{\text{id}}$. The state bound $|x_k| \leq 0.75$ is softened by the slack variables in (48) and may therefore be violated; the input bound is hard.

After the initialization phase, the reference is an amplitude-swept sinusoid. For the j th segment, the reference is defined by

$$r_k = A_j \sin(2\pi f(k - N_{\text{id}})T_s),$$

where $f = 0.10$ Hz and

$$A_j \in \{0.25, 0.55, 0.80, 0.90, 0\}.$$

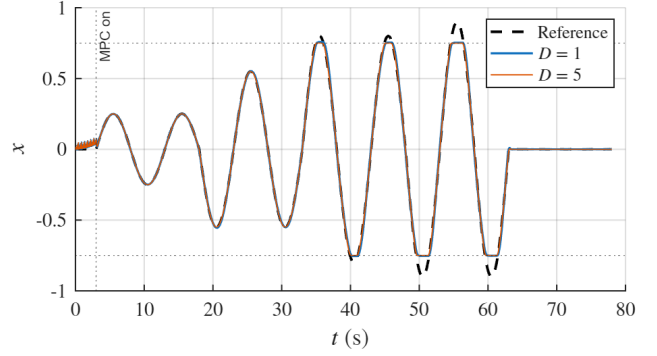


Fig. 1. Closed-loop state tracking for the first- and fifth-order Taylor models. The horizontal dotted lines indicate the nominal bound $|x| \leq 0.75$, which is softened with slack variables and may be violated.

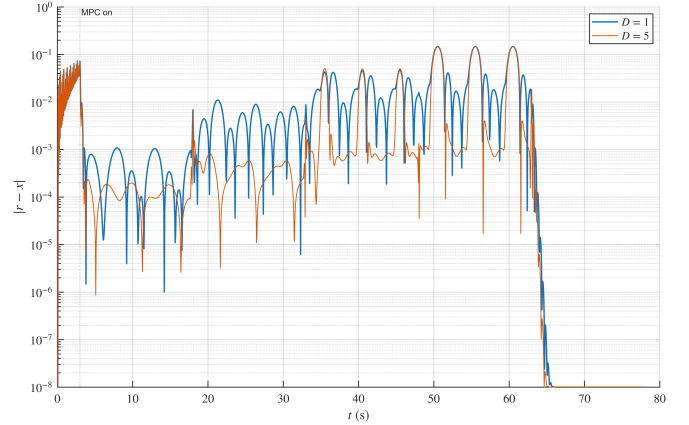


Fig. 2. Absolute tracking error on a logarithmic vertical scale for the first- and fifth-order Taylor models. The plot includes the segment transients.

Each amplitude is maintained for $K_r = 300$ samples before transitioning to the next level. The increasing amplitudes progressively excite higher-order nonlinearities. Because the state constraint is soft, the closed-loop state can exceed its nominal bound.

The cases are compared through state and input trajectories, tracking errors, and quantitative performance metrics computed over each amplitude segment.

B. Closed-Loop Results

Figures 1–3 summarize the closed-loop responses produced by the proposed controller. Figures 1 and 2 show the state trajectories and logarithmic tracking errors, respectively.

Each per-segment metric is computed over its 300-sample half-open interval $[\text{segmentStart}, \text{segmentStop})$. For the evaluated post-transient MAE in Figure 3, the first 200 samples are discarded and the remaining 100 are averaged. In contrast, the overall RMSE and MAE in Table III use all stored samples $k \geq N_{\text{id}}$, including the final sample $k = K$.

RMSE and MAE measure overall and average tracking error, respectively. The maximum state violation is the largest exceedance of the nominal soft state bound; total input variation (I.V.) quantifies accumulated control variation.

Figure 1 compares the first- and fifth-order Taylor models. Both responses track the reference, with visible clipping near the softened state bound during the larger-amplitude segments. The nonzero violations in Table III confirm that

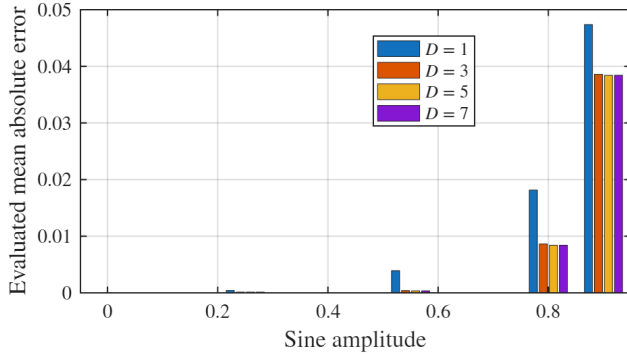


Fig. 3. Evaluated post-transient mean absolute error (MAE) for each reference amplitude and Taylor degree. For every 300-sample amplitude segment, the first sinusoidal period (200 samples) is discarded and the MAE is evaluated over the remaining 100 samples.

TABLE III
CLOSED-LOOP OVERALL PERFORMANCE FOR DIFFERENT TAYLOR DEGREES D .

D	RMSE	MAE	Maximum state violation	Total input variation
1	0.03199	0.01418	7.37×10^{-3}	12.41
3	0.03092	0.00971	3.50×10^{-4}	11.75
5	0.03090	0.00958	2.56×10^{-4}	11.86
7	0.03089	0.00959	3.52×10^{-4}	11.81

exact state-constraint satisfaction is not claimed.

Figure 2 presents the corresponding absolute tracking errors, including the segment transients, on a logarithmic vertical scale. This scale displays the small within-segment errors and the larger error peaks in the same panel.

Figure 3 quantifies this trend through the evaluated post-transient MAE for each sinusoidal amplitude after the 200-sample discard. This metric is distinct from the overall MAE in Table III. Differences among Taylor degrees are negligible at small amplitudes. As the amplitude increases, the first-order model shows consistently larger mean errors than the higher-order models, while the third-, fifth-, and seventh-order models remain close to one another. This behavior is consistent with higher-order nonlinear terms becoming more influential as the system operates farther from the equilibrium about which the Taylor expansion is constructed.

IX. CONCLUSION

This paper presented a Taylor-informed indirect adaptive predictive control framework for nonlinear systems based on Jacobian-frozen affine predictors. In the numerical study, the continuous-time plant is simulated under zero-order hold using RK4 with $T_s = 0.05$ s, and RLS identifies the resulting one-step transitions. Online identification of the implemented forward-Euler/Taylor-structure-informed reduced dictionary yields an adaptive sampled nonlinear map. At each sampling instant, its Jacobian is evaluated at the current operating point and frozen over the prediction horizon, producing an affine predictor for finite-horizon model predictive control.

The results show lower overall MAE for $D \in \{3, 5, 7\}$ than for $D = 1$, with similar total input variation. Joint-odd symmetry exactly removes even-total-degree monomials. Deliberate forward-Euler-informed pruning reduces the model

but does not imply that omitted monomials vanish in the RK4 sampled flow. Maximum soft-state violations are reported.

Future work will investigate variable-rate forgetting for improved adaptation under changing operating conditions, extend the framework to systems with time delays [18], and validate the approach on real-world physical platforms.

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REFERENCES

- [1] D. W. Clarke, C. Mohtadi, and P. S. Tuffs, "Generalized predictive control—part I: The basic algorithm," *Automatica*, vol. 23, no. 2, pp. 137–148, 1987.
- [2] —, "Generalized predictive control—part II: Extensions and interpretations," *Automatica*, vol. 23, no. 2, pp. 149–160, 1987.
- [3] J. B. Rawlings, D. Q. Mayne, and M. Diehl, *Model Predictive Control: Theory, Computation, and Design*. Nob Hill Publishing, 2020, vol. 2.
- [4] R. J. Richards, S. A. U. Islam, and D. S. Bernstein, "Predictive cost adaptive control of the NASA benchmark flutter model," *Journal of Guidance, Control, and Dynamics*, vol. 48, no. 12, pp. 2663–2679, 2025.
- [5] T. W. Nguyen, "Fast RLS identification leveraging the linearized system sparsity: Predictive cost adaptive control for quadrotors," in *2026 American Control Conference (ACC)*. IEEE, 2026.
- [6] J. C. Vander Schaaf, K. J. Fidkowski, and D. S. Bernstein, "Active flow control using adaptive model predictive control with online, closed-loop system identification," *Journal of Guidance, Control, and Dynamics*, vol. 49, no. 7, pp. 1883–1896, 2026.
- [7] F. Allgöwer and A. Zheng, *Nonlinear Model Predictive Control*. Birkhäuser, 2012.
- [8] M. Kamalidar, "Iterative state- and control-dependent model predictive control: A jacobian-free formulation for constrained nonlinear systems," 2026. [Online]. Available: <https://arxiv.org/abs/2608.15322>
- [9] R. Tao, P. Zhao, I. Kolmanovsky, and N. Hovakimyan, "Robust adaptive MPC in the presence of nonlinear time-varying uncertainties: An uncertainty compensation approach," *Automatica*, vol. 192, p. 113142, 2026.
- [10] A. Burtsev, A. Jariwala, E. Bakolas, and D. Goldstein, "Adaptive feedback flow control for wings," *AIAA Journal*, pp. 1–14, 2026.
- [11] M. Castroviejo-Fernandez, H.-T. Do, and I. Kolmanovsky, "Constraint representation through support vector machines and its application to model predictive control," *Engineering Applications of Artificial Intelligence*, vol. 181, p. 115444, 2026.
- [12] T. W. Nguyen, S. A. U. Islam, D. S. Bernstein, and I. V. Kolmanovsky, "Predictive cost adaptive control: A numerical investigation of persistency, consistency, and exigency," *IEEE Control Systems Magazine*, vol. 41, no. 6, pp. 64–96, 2021.
- [13] T. W. Nguyen, "Adaptive behavioral predictive control: State-free regulation without hankel weights," 2026. [Online]. Available: <https://arxiv.org/abs/2602.12016>
- [14] R. A. Alhazmi, A. S. Babu, S. A. U. Islam, and D. S. Bernstein, "Nonlinear predictive cost adaptive control of pseudo-linear input-output models using polynomial, Fourier, and cubic spline observables," 2026. [Online]. Available: <https://arxiv.org/abs/2602.05263>
- [15] Y. Engel, S. Mannor, and R. Meir, "The kernel recursive least-squares algorithm," *IEEE Transactions on Signal Processing*, vol. 52, no. 8, pp. 2275–2285, 2004.
- [16] W. Liu, I. Park, Y. Wang, and J. C. Principe, "Extended kernel recursive least squares algorithm," *IEEE Transactions on Signal Processing*, vol. 57, no. 10, pp. 3801–3814, 2009.
- [17] J. A. Rosenfeld, B. P. Russo, R. Kamalapurkar, and T. T. Johnson, "The occupation kernel method for nonlinear system identification," *SIAM Journal on Control and Optimization*, vol. 62, no. 3, pp. 1643–1668, 2024.
- [18] W. Michiels and B. Zhou, "Delay compensation using predictive and periodic feedback," in *Nonlinear and Constrained Control: Applications, Synergies, Challenges and Opportunities*, E. Garone, I. Kolmanovsky, and T. W. Nguyen, Eds. Cham: Springer Nature Switzerland, 2025, pp. 249–270.